DATA-DRIVEN WORKFORCE INTELLIGENCE

Job Market Analysis

Trends · Demand · Skills · Salaries · Strategy

PROJECT OBJECTIVE

Objective

Analyze the job market for Data Science and Analytics roles using 742 real-world job postings to identify hiring trends, salary patterns, industry demand, company hiring behavior, skill requirements, and educational influences on compensation.

Five Core Focus Areas

- Hiring Demand
- Salary Patterns
- Skill Requirements
- Education Impact
- Market Trends

Hiring Trends

Identify which locations, industries, and companies are hiring the most

Salary Patterns

Uncover compensation trends by role, education, and experience level.

Education Impact

Degree Level vs. Pay correlation

DATASET OVERVIEW

Dataset Overview

742
Job Postings

343
Companies

60
Industries

DATA COVERAGE

- Job Titles
- Salaries
- Locations
- Industries
- Technical Skills
- Companies
- Education Levels

BUSINESS VALUES

- Combines job market demand, salary information, organizational characteristics, and skill requirements
- Supports employers in benchmarking compensation
- Helps job seekers identify high-demand skills
- Enables workforce planners to track hiring trends
- Provides research value for HR analytics and academic projects

Data Cleaning & Preprocessing

Missing Value Assessment

Identified and reviewed all fields with null or missing entries across the dataset.

Duplicate Record Removal

Detected and removed duplicate records to maintain dataset integrity.

Categorical Field Validation

Ensured categorical columns such as Degree, Industry, and Location contained valid entries.

Job Title Standardization

Standardized job title variations using the Jobtitle_sim field for consistent grouping.

Quality Assurance Checks

Performed a final QA pass across all major columns before proceeding with analysis.

Salary Field Validation

Verified that Lower_Salary, Upper_Salary, and Avg_SalaryK fields were logically consistent.

ANALYSIS METHODS

Analysis Methodology

01

SQL Analysis

Used structured queries to extract, filter, aggregate, and rank data across key dimensions such as location, industry, job title, salary, and skills.

02

Dashboard Analysis

Built an interactive dashboard providing a visual summary of hiring trends, salary patterns, and skill distribution across all job postings.

03

KPI Analysis

Monitored four core KPIs: Total Job Postings (742), Average Salary, Total Companies (343), and Total Industries (60).

04

Comparative Analysis

Compared variables across dimensions — skills vs. salary, education vs. salary, experience vs. salary — to uncover actionable patterns.

05

Business Interpretation

Each finding was translated into practical implications for employers, job seekers, and workforce planners.

SQL Queries Analysis and Insights

Purpose : Identify which U.S. states have the highest number of data science job postings.

```
SELECT Job_Title, COUNT(*) AS Total_Jobs
FROM Market GROUP BY Job_Title
ORDER BY Total_Jobs DESC LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Job_Title	Total_Jobs
Data Scientist	131
Data Engineer	53
Senior Data Scientist	34
Data Analyst	15
Senior Data Engineer	14
Senior Data Analyst	12
Lead Data Scientist	8
Sr. Data Engineer	6
Marketing Data Analyst	6
Principal Data Scientist	5

KEY FINDINGS

- Reveals the top 10 states by job count
- Highlights geographic concentration of opportunities
- Helps job seekers prioritize relocation decisions

BUSINESS INTERPRETATION

States with the most postings represent the strongest hiring markets. Job seekers benefit by targeting these regions, while employers can benchmark hiring activity against peers.

SQL Queries Analysis and Insights

Purpose : Compare average minimum and maximum salary ranges across states.

```
SELECT RIGHT(Location, 2),
ROUND(AVG(Lower_Salary),2) AS Avg_Min_Salary,
ROUND(AVG(Upper_Salary),2) AS Avg_Max_Salary
FROM Market GROUP BY RIGHT(Location, 2)
ORDER BY Avg_Max_Salary DESC LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

State	Avg_Min_Salary	Avg_Max_Salary
CA	92.43	154.60
IL	88.35	144.98
MA	78.69	136.30
NJ	76.82	132.29
DC	88.64	131.73
NC	69.05	127.86
KY	66.33	127.67
MD	68.31	126.40
RI	74.00	126.00
NY	72.13	125.18

KEY FINDINGS

- Shows salary ceiling and floor by state
- Reveals which states offer the widest salary ranges
- Useful for compensation benchmarking

BUSINESS INTERPRETATION

States with high salary ceilings attract top talent. Employers must offer competitive upper-range salaries to remain attractive in high-cost markets.

SQL Queries Analysis and Insights

Purpose : Rank states by their average data science compensation.

```
SELECT RIGHT(Location, 2),
ROUND(AVG(Avg_SalaryK),2) AS Avg_Salary
FROM Market GROUP BY RIGHT(Location, 2)
ORDER BY Avg_Salary DESC LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

State	Avg_Salary
CA	123.51
IL	116.66
DC	110.18
MA	107.50
NJ	104.56
MI	100.25
RI	100.00
NY	98.65
NC	98.45
MD	97.36

KEY FINDINGS

- Ranks top 10 states by average salary
- Bellevue, WA leads with avg salary of $184.5K
- Salary varies significantly across regions

BUSINESS INTERPRETATION

Location has a major impact on compensation. Professionals aiming for maximum earnings should consider relocating to high-salary states such as Washington and California.

INSIGHTS FROM SQL QUERY ANALYSIS

SQL Queries Analysis and Insights

Purpose : Identify which industries have the highest demand for data science roles.

```sql
SELECT Industry,
COUNT(*) AS Total_Jobs
FROM Market WHERE Industry <> '-1' GROUP BY Industry
ORDER BY Total_Jobs DESC LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Industry	Total_Jobs
Biotech & Pharmaceuticals	112
Insurance Carriers	63
Computer Hardware & Software	59
IT Services	50
Health Care Services & Hospitals	49

KEY FINDINGS

- Biotech & Pharma leads with 112 postings
- IT and Finance sectors follow closely
- Healthcare dominates analytics hiring

BUSINESS INTERPRETATION

Industries with the most postings are investing heavily in analytics. Job seekers should focus skills toward Healthcare and Biotech where demand is highest and talent is scarce.

SQL Queries Analysis and Insights

Purpose : Identify which companies are the most active recruiters for data roles.

```
SELECT Company_Name, COUNT(*) AS Total_Openings
FROM Market GROUP BY Company_Name
ORDER BY Total_Openings DESC LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Company_Name	Total_Openings
MassMutual 3.6	14
Reynolds American 3.1	14
Takeda Pharmaceuticals 3.7	14
Software Engineering Institut...	11
Liberty Mutual Insurance 3.3	10
PNNL 3.8	10
AstraZeneca 3.9	9
MITRE 3.2	8
Rochester Regional Health 3.3	7
Fareportal 3.8	7

KEY FINDINGS

- Top 10 companies account for significant share of postings.
- Large enterprises dominate hiring activity.
- Reveals key target employers for job seekers.

BUSINESS INTERPRETATION

Companies hiring repeatedly are expanding analytics capabilities. Job seekers should target these organizations for better prospects, while recruiters can benchmark hiring volumes against market leaders.

SQL Queries Analysis and Insights

Purpose : Determine which job roles have the highest market demand.

```
SELECT Job_Title, COUNT(*) AS Total_Jobs
FROM Market GROUP BY Job_Title
ORDER BY Total_Jobs DESC LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Job_Title	Total_Jobs	Avg_Salary
Data Scientist	131	106.18
Data Engineer	53	91.66
Senior Data Scientist	34	134.87
Data Analyst	15	66.30
Senior Data Engineer	14	121.93
Senior Data Analyst	12	83.42
Lead Data Scientist	8	161.25
Sr. Data Engineer	6	115.00
Marketing Data Analyst	6	48.67
Machine Learning Engineer	5	104.90

KEY FINDINGS

- Data Scientist is the #1 role with 131 postings
- Data Engineer has 53 postings
- Demand for DS roles far exceeds other titles

BUSINESS INTERPRETATION

The dominance of Data Scientist roles reflects the industry shift toward AI and ML. Professionals developing these skills can expect broader and better-paying career opportunities.

SQL Queries Analysis and Insights

Purpose : Combine job title demand with average salary to assess value in the market.

```
SELECT Job_Title, COUNT(*) AS Total_Jobs,
ROUND(AVG(Avg_SalaryK),2) AS Avg_Salary
FROM Market GROUP BY Job_Title
ORDER BY Total_Jobs DESC LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Job_Title	Total_Jobs	Avg_Salary
Data Scientist	131	106.18
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Lead Data Scientist	8	161.25
Sr. Data Engineer	6	115.00
Marketing Data Analyst	6	48.67
Machine Learning Engineer	5	104.90

KEY FINDINGS

- Lead Data Scientist earns avg $161.25K
- Data Analyst earns avg $66.30K
- Senior roles offer 2-3x the compensation

BUSINESS INTERPRETATION

Salary gaps between junior and senior roles are substantial. Investing in career advancement and specialization delivers strong financial returns for analytics professionals.

SQL Queries Analysis and Insights

Purpose : Map required technical skills to each job role category.

```
SELECT job_title_sim,
SUM(Python) AS Python, SUM(sql_) AS SQL_Skill,
SUM(excel) AS Excel, SUM(tableau) AS Tableau,
SUM(aws) AS AWS, SUM(spark) AS Spark,
SUM(hadoop) AS Hadoop, SUM(sas) AS SAS,
SUM(scikit) AS Scikit, SUM(pytorch) AS PyTorch,
SUM(keras) AS Keras, SUM(mongo) AS MongoDB
FROM Market GROUP BY job_title_sim
ORDER BY job_title_sim;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

job_title_sim	Python	SQL_Skill	Excel	Tableau	AWS	Spark	Hadoop	SAS	Scikit	PyTorch	Keras	MongoDB
analyst	31	75	77	40	10	6	3	11	0	0	0	3
data analitics	5	4	4	4	0	0	0	2	0	0	0	0
data engineer	77	87	54	11	59	67	50	1	0	0	0	13
data modeler	2	4	4	0	2	1	1	0	0	0	0	0
data scientist	240	176	155	76	79	84	60	50	47	33	29	19
Data scientist project manager	4	11	11	12	2	0	0	0	0	0	0	0
director	0	0	2	0	1	0	0	0	0	0	0	0
machine learning engineer	18	13	7	0	6	4	4	0	7	5	0	0
na	4	7	9	5	3	5	6	0	0	0	0	2
other scientist	11	3	65	0	14	0	0	2	0	1	0	0

BUSINESS INTERPRETATION

Skill requirements differ significantly by role. Candidates should tailor their skill development to match the specific technical stack required by their target position.

KEY FINDINGS

- Python and SQL appear most frequently across all roles
- AWS and Spark are key for Data Engineering roles
- Tableau and Excel dominate Analyst roles

SQL Queries Analysis and Insights

Purpose : Determine how educational qualifications affect compensation.

```
SELECT Degree,
ROUND(AVG(Avg_SalaryK),2) AS Average_Salary,
COUNT(*) AS Total_Jobs
FROM Market WHERE Degree <> '-1' GROUP BY Degree
ORDER BY Average_Salary DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Degree	Average_Salary	Total_Jobs
P	115.98	107
M	105.63	252
na	94.70	383

KEY FINDINGS

- PhD holders earn avg $115.98K
- Master's degree earns avg $105.63K
- Higher education clearly correlates with higher pay

BUSINESS INTERPRETATION

Advanced education delivers a measurable salary premium. Professionals targeting senior or research-oriented roles should consider pursuing higher qualifications or specialized certifications. (M = Master's | P = PhD | na = None)

SQL KEY FINDINGS

SQL Insights

Job Roles

Data Scientist (131 openings) is the most demanded role, more than double that of Data Engineers (53).

Location

New York, California, and Washington lead in job postings; Bellevue, WA offers the highest average salary at $184.5K.

Industry

Biotech & Pharmaceuticals (112 postings) and IT are the top hiring industries, reflecting strong analytics adoption.

Core Skills

Python and SQL appear across all roles; AWS and Spark are critical for Data Engineering; Tableau/Excel for Analysts.

Companies

Large enterprises drive the majority of openings; top 10 companies account for a disproportionate share of postings.

Education ↑ Pay

PhD roles average $115.98K; Master's roles average $105.63K — higher education drives higher compensation.

Cross – Analysis of key Variables

SKILL VS SALARY

Finding: Roles requiring Python, SQL, AWS, and Spark offer higher average salaries.
Business: Professionals with multi-skill portfolios attract better compensation.

INDUSTRY VS HIRING

Finding: Healthcare & Biotech lead hiring, driven by data-driven decision-making needs.
Business: Targeting these sectors increases employment probability for analytics talent.

LOCATION VS OPPORTUNITIES

Finding: Cities with the most job postings offer the best career opportunities.
Business: Candidates should weigh both salary and cost of living before relocating.

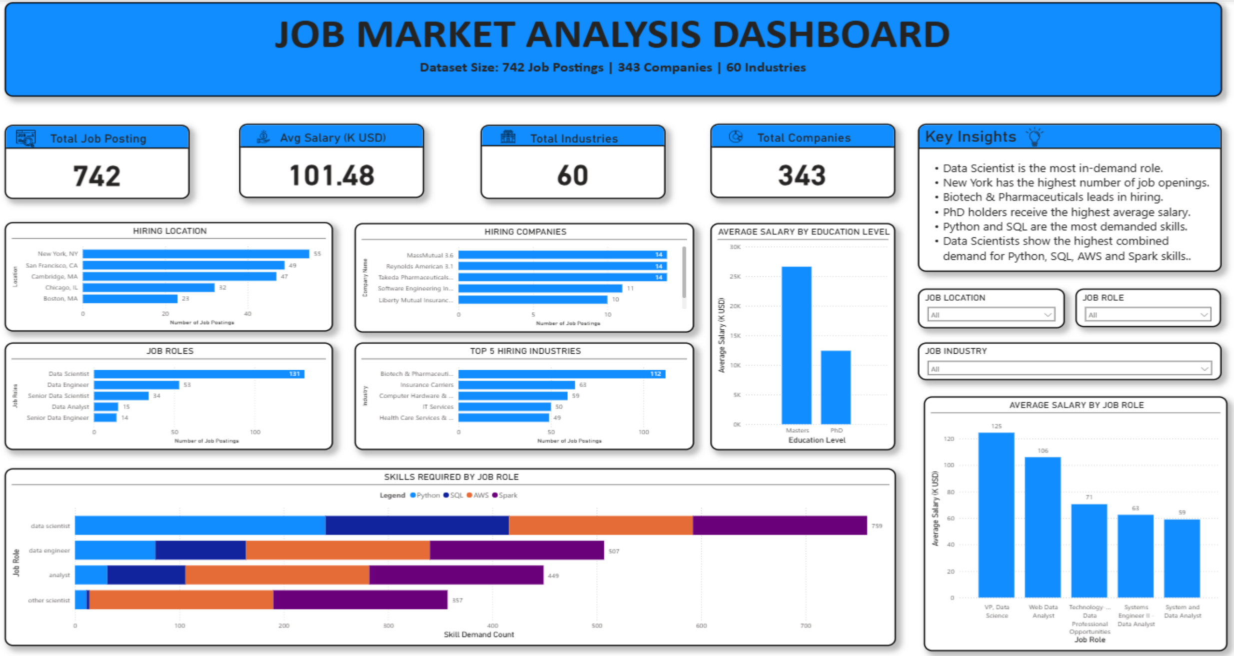
EDUCATION VS SALARY

Finding: Higher education qualifications are associated with higher average salaries.
Business: Advanced degrees provide an advantage for specialized and senior positions.

Dashboard Overview

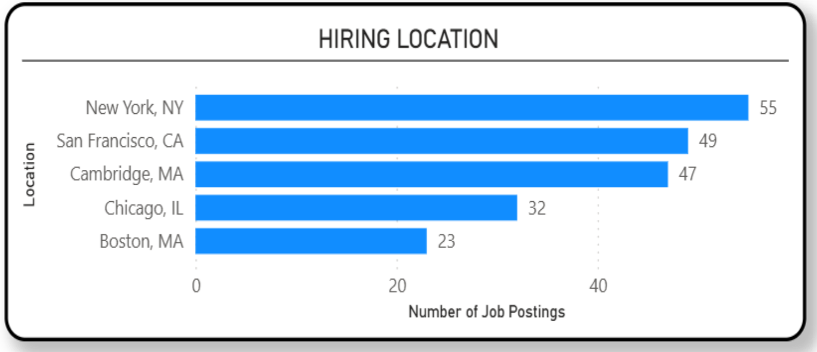
Dashboard Overview

The dashboard provides an integrated view of job market trends, including hiring locations, industries, companies, salaries, education requirements, and skill demand. It serves as a decision-support tool for analyzing workforce patterns and identifying key market opportunities.



Dashboard Charts

Hiring by Location



KEY OBSERVATION

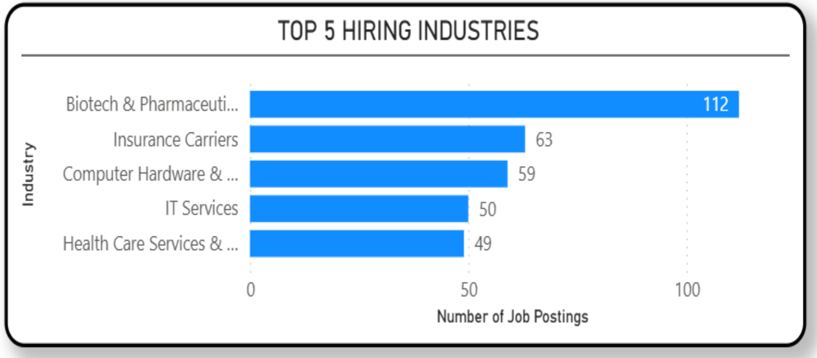
New York and California dominate the map with the highest number of job postings. This geographic concentration reflects where analytics-intensive industries are most active.

BUSINESS INTERPRETATION

This suggests New York is a major analytics hub. Job seekers targeting maximum opportunity should consider major coastal cities and technology corridors.

Dashboard Charts

Hiring Industries



KEY OBSERVATION

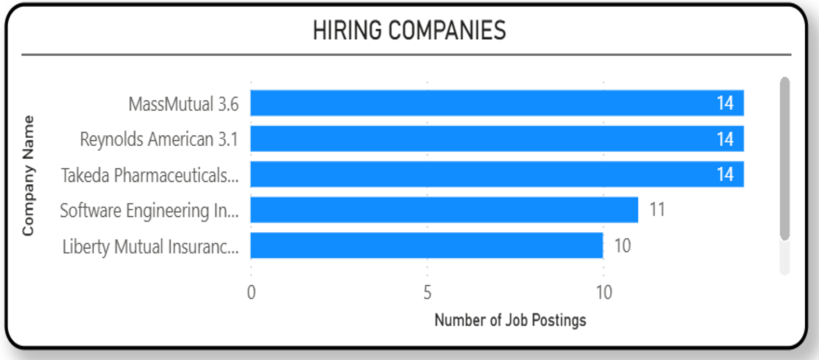
Biotech & Pharmaceuticals (112 postings) leads all industries. IT, Finance, and Insurance follow, collectively representing the majority of job postings.

BUSINESS INTERPRETATION

Healthcare & Biotechnology's dominance reflects increasing investment in data-driven healthcare and research. Professionals with domain knowledge in these fields command higher demand.

Dashboard Charts

Hiring Companies



KEY OBSERVATION

Large enterprises and technology firms hold the highest hiring volumes. The top 10 companies account for a significantly disproportionate share of total postings.

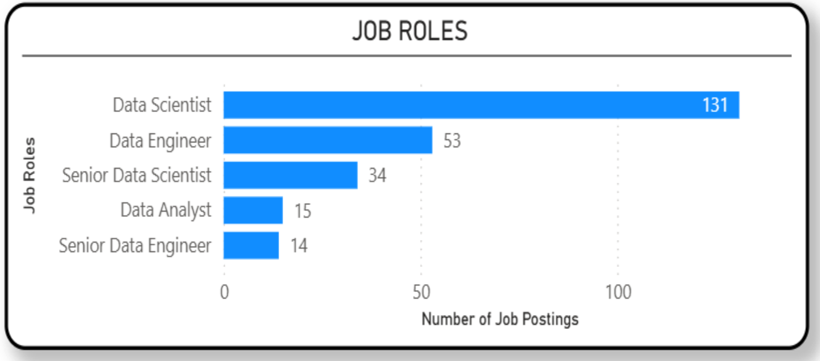
BUSINESS INTERPRETATION

Companies hiring repeatedly are likely expanding analytics capabilities. Job seekers should target these organizations for better career advancement prospects and resources.

INTERACTIVE DASHBOARD

Dashboard Charts

Job Role Distribution



KEY OBSERVATION

Data Scientist dominates with 131 openings. Data Engineers (53), Analysts, and Machine Learning Engineers follow. Entry-level positions are proportionally lower.

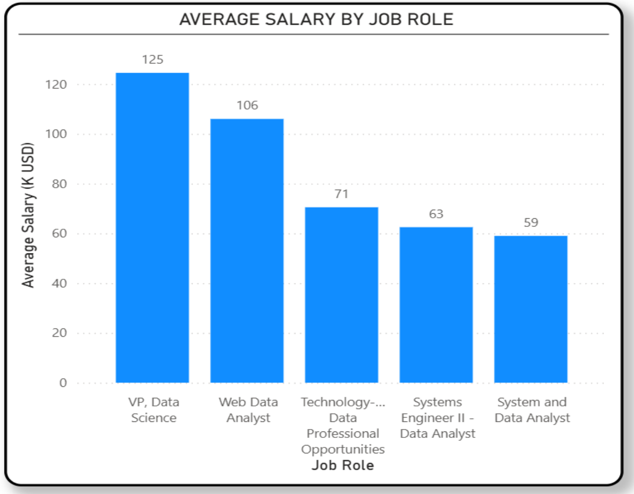
BUSINESS INTERPRETATION

The high demand for Data Scientists and Data Engineers highlights increasing focus on predictive analytics and ML. Candidates preparing for these roles can expect broader opportunities.

INTERACTIVE DASHBOARD

Dashboard Charts

Average Salary by Job Role



KEY OBSERVATION

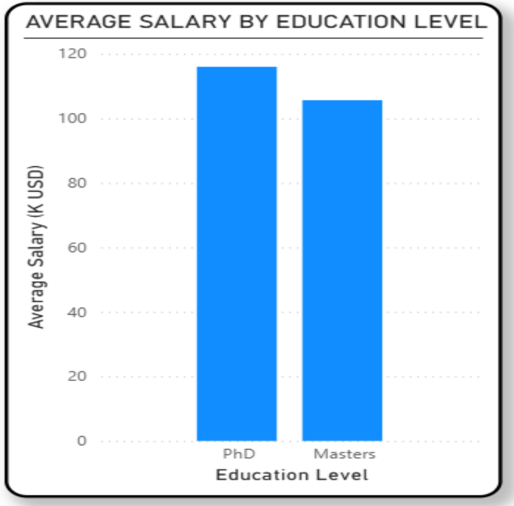
Lead Data Scientist earns an average of $161.25K. Director and Principal roles follow at similar levels. Data Analysts earn the lowest average salary at $66.30K.

BUSINESS INTERPRETATION

Specialized and senior analytical roles attract higher salaries due to broader technical expertise and greater business responsibility. Career progression toward senior roles yields substantial financial rewards.

Dashboard Charts

Average Salary by Education Level



KEY OBSERVATION

PhD-required roles average $115.98K. Master's degree roles average $105.63K. Roles with no degree requirement offer the lowest average compensation.

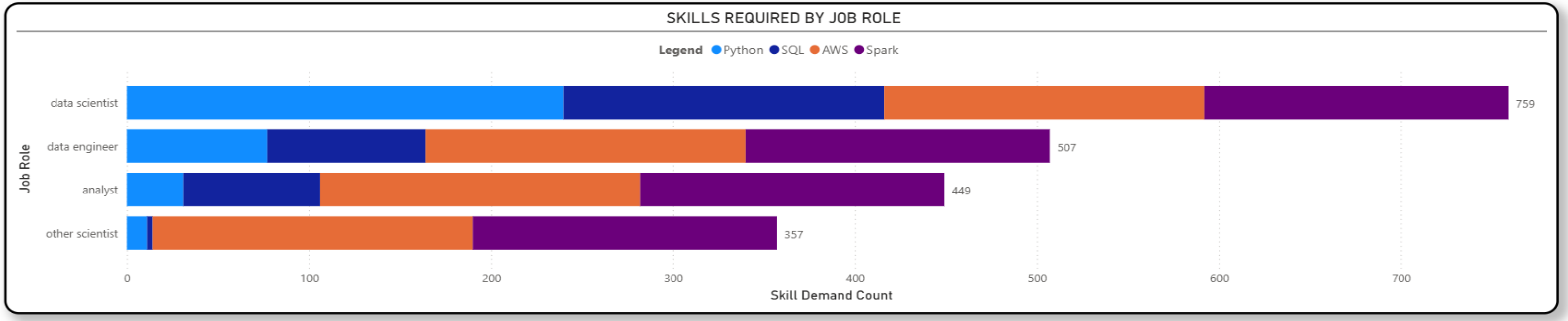
BUSINESS INTERPRETATION

Higher education provides a measurable salary advantage. While practical skills are critical, advanced degrees open doors to specialized and higher-paying positions in analytics and research.

INTERACTIVE DASHBOARD

Dashboard Charts

Average Salary by Education Level



KEY OBSERVATION

Python and SQL appear consistently across all job roles. AWS and Spark are most prominent for Data Engineers. Tableau and Excel are key for Analyst roles. ML frame works (Keras, PyTorch) are critical for Data Scientists.

BUSINESS INTERPRETATION

Python, SQL, AWS, and Spark frequently appear together, showing employers increasingly expect a combination of programming, cloud, and big-data skills for advanced analytics roles.

KEY FINDINGS

Business Insights

Highest Demand Role

Data Scientist is the #1 role with 131 openings. Organizations are prioritizing DS to support AI initiatives and data-driven decision-making.

Location Matters

Bellevue, WA leads salary at $184.5K. Geographic choices significantly impact earning potential and career opportunities.

Healthcare Leads Hiring

Biotech & Pharma generated 112 postings — the largest hiring industry. Analytics is transforming healthcare and research.

Core Skills

Python is the most-requested skill across DS, DE, and Analyst roles. SQL appears consistently across every job category.

Cloud & Big Data Skills

AWS, Spark, and Hadoop are valuable differentiators, especially for Data Engineers and senior analytics roles.

Education Impacts Salary

PhD roles earn $115.98K; Master's $105.63K. Advanced education correlates with higher pay and specialized roles.

STRATEGIC RECOMMENDATIONS

Recommendations : For Job Seekers

01

Master Python & SQL

These two skills are consistently the most demanded technical competencies across all data roles. They form the foundation of any data career.

02

Learn Cloud & Big Data

Develop expertise in AWS and big data technologies like Spark to remain competitive as organizations scale their data infrastructure.

03

Build a Portfolio

Create practical projects demonstrating real-world analytical and technical capabilities. Employers value demonstrated experience alongside credentials.

04

Target High-Growth Industries

Focus on Healthcare & Biotechnology and similar sectors where hiring demand is significantly above average for analytics talent.

05

Consider Geographic Strategy

Explore opportunities in major technology hubs like New York, California, and Washington where hiring density and salary levels are highest.

CAREER STRATEGY

Recommendations : For Employers

01

Invest in Upskilling

Launch employee development programs focused on cloud computing and modern data technologies to close the growing skills gap internally.

02

Offer Competitive Packages

Provide competitive compensation to attract skilled analytics professionals, particularly for senior and specialized roles commanding $100K+.

03

Value Both Experience & Education

Recruit based on demonstrated practical project experience and academic qualifications together — neither alone is sufficient.

04

Diversify Talent Sourcing

Expand hiring strategies beyond traditional talent pools to address increasing demand and reduce dependence on a narrow candidate pipeline.

05

Encourage Continuous Learning

Implement certification programs and learning stipends to improve workforce capabilities and reduce employee attrition in competitive markets.

TALENT STRATEGY

CONCLUSION

Strategic Workforce Readiness

Market Intelligence

Skill Gap Closure

Hiring Optimization

The project successfully converted 742 raw job market postings into a comprehensive, actionable analytical framework. Every dimension of the job market — hiring demand, salary trends, industry opportunities, educational impact, and skill requirements — was systematically examined.